

Flood Impact Assessment and Dynamic Mapping

Factor	Dataset Needed	Where to Request in Sri Lanka	Format	Resolution	Purpose
Historical Flood Records	Past flood event records	Disaster Management Centre (dmc.gov.lk); DesInventar Sri Lanka (desinventar.net, country: LKA)	Tabular	Event-level	Historical severity benchmarking and model training data
River Water Level Data	River gauge water level readings	Irrigation Department of Sri Lanka (irrigation.gov.lk); Department of Hydrology; River Gauge Stations	Time-series (CSV/API)	Hourly / daily, per gauge station	Real-time flood threshold detection and rising-water alerts
Rainfall Data	Rain gauge network + historical rainfall records	Department of Meteorology (meteo.gov.lk)	Time-series (CSV/API)	Hourly / daily, per station	Primary trigger input for flood forecasting model
Digital Elevation Model (DEM)	Elevation / terrain model	Survey Department of Sri Lanka (survey.gov.lk); SRTM or ALOS PALSAR if unavailable locally	Raster (GeoTIFF)	<10m (requested); 30m (SRTM) / 12.5m (ALOS) fallback	Flood extent modeling, flow accumulation, and low-lying area identification
Land Use / Land Cover Data	Land use classification	Survey Department of Sri Lanka (survey.gov.lk); Ministry of Environment (env.gov.lk)	Raster / Vector	10m or 1:50,000 base	Surface runoff estimation; impervious surface and vulnerable land use identification
Satellite Imagery (Flood Extent)	Flood extent detection imagery	Copernicus Sentinel-1 & Sentinel-2 (ESA – browser.dataspace.copernicus.eu); USGS Landsat (earthexplorer.usgs.gov)	Raster (SAR / optical)	10m (Sentinel) / 30m (Landsat)	Actual flood extent mapping and validation of model predictions
Administrative Boundaries (Province, District, DS, GN)	Administrative boundary layers	Survey Department of Sri Lanka (survey.gov.lk); Department of Census & Statistics (statistics.gov.lk)	Vector (polygon)	Province / District / DS / GN division level	Spatial aggregation unit for combining all component datasets and reporting
River Network / Drainage Data	River and drainage network layers	Survey Department of Sri Lanka (survey.gov.lk); Irrigation Department (irrigation.gov.lk)	Vector (line)	1:50,000 topo base	Hydrological network modeling and downstream flood propagation
Historical Damage & Loss Records	Damage and loss records from past floods	Disaster Management Centre (dmc.gov.lk); DesInventar Sri Lanka (desinventar.net, country: LKA)	Tabular	Event / district level	Loss estimation, impact severity modeling, and damage benchmarking

Probabilistic Assessment and Forecasting of Landslide Occurrences using ML

Factor	Dataset Needed	Where to Request in Sri Lanka	Format	Resolution	Purpose
Slope Angle	DEM (LiDAR / high-resolution elevation)	Survey Department – Geoportal (ilas.survey.gov.lk/geoportal) or Map Portal (app.survey.gov.lk/mapportal). Higher-res DEM via Geoinformatics Branch: dsgeoinfo@survey.gov.lk	Raster (GeoTIFF)	<10m (requested); 30m fallback	Core susceptibility input; maps to NBRO risk bands (>35° Very High → 11–16° Low)
Soil Classification	Soil type maps (Red-Yellow Podzolic / Mountain Regosol)	Department of Agriculture (doa.gov.lk) and NBRO LRRMD geotechnical reports (info@nbro.gov.lk)	Vector (polygon)	District / mountain-block level	Identifies clay-trap zones (Badulla, Nuwara Eliya, Kandy) vs colluvium zones (Kegalle, Ratnapura)
Geology / Bedrock Type	Bedrock and weathering-depth maps	Geological Survey & Mines Bureau (GSMB) – request via office, gsmb.gov.lk	Vector (polygon)	District level	Structural weakness / deep weathering (e.g. fragmented granitic gneiss at Aranayaka)
Land Use / Land Cover	Land use classification, forest cover	Survey Department – topographic vector databases (survey.gov.lk)	Raster / Vector	10m or 1:50,000 base	Deforestation proxy – loss of root-anchoring correlates with failure
Hydrology / Drainage Network	Stream network, drainage lines	Survey Department – topographic data layers (survey.gov.lk)	Vector (line)	1:50,000 topo base	Proximity to drainage; saturation and channel-adjacent risk
Rainfall Intensity (real-time)	Rain gauge network + historical rainfall records	Department of Meteorology (meteo.gov.lk); NBRO automated rain gauges via LRRMD	Time-series (CSV/API)	Hourly / daily, per station	NBRO trigger thresholds: 75mm/24h Watch, 100mm/24h Alert, 150mm/24h or 75mm/1h Evacuation
Soil Saturation / Moisture	Soil moisture readings or piezometer data	NBRO LRRMD (instrumented sites) – info@nbro.gov.lk	Sensor time-series	Point-level, per site	Captures antecedent moisture and pore-water pressure buildup
Dangerous Slope Cutting / Building Footprint	Building locations, construction permits	Survey Department building layer (survey.gov.lk); local Divisional Secretariat / local authority permits	Vector (point/polygon)	Building-level	Flags artificial slope destabilization from excavation or construction
Population Exposure	Population counts by administrative division	Department of Census & Statistics (statistics.gov.lk)	Vector / tabular	Grama Niladhari division level	Weights risk score by number of people actually exposed
Infrastructure Accessibility	Road network data	Survey Department (survey.gov.lk); Road Development Authority (RDA)	Vector (line)	Road-segment level	Evacuation route viability and emergency-response prioritization
Historical Landslide Inventory / Hazard Zones	Landslide hazard zonation maps, past event records	NBRO LRRMD data portal (liris.nbro.lk/en); email info@nbro.gov.lk, subject: "Attn: Director, LRRMD - Landslide Dataset Request"	Vector polygon (hazard_type, zone_id, risk_score)	Zone level	Ground-truth labels for training and validating the ML model

Early Warning Decision Support Component

Factor	Dataset Needed	Where to Request in Sri Lanka	Format	Resolution / Spatial Level	Purpose
Flood Hazard	Flood prediction outputs (probability, depth, extent, risk level, time)	Team Flood Prediction Model / Department of Irrigation (www.irrigation.gov.lk)	Raster / GeoJSON / CSV	GN Division / Grid level (~100m–250m)	Primary flood hazard input for joint hazard analysis and priority scoring
Landslide Hazard	Landslide prediction outputs (probability, susceptibility, risk level, time)	Team Landslide Prediction Model / National Building Research Organisation (NBRO) – (www.nbro.gov.lk)	Raster / GeoJSON / CSV	GN Division / Grid level (~100m–250m)	Primary landslide hazard input for joint hazard analysis and priority scoring
Rainfall Trigger	Rain gauge observations and historical rainfall data (intensity, duration)	Department of Meteorology (www.meteo.gov.lk)	CSV / API (Time-series)	Hourly / Daily per station	Identifies heavy rainfall events triggering floods and landslides; used for validation
Digital Elevation Model (DEM)	High-resolution elevation data (for slope, aspect, height)	Survey Department of Sri Lanka (www.survey.gov.lk) or NASA SRTM / Copernicus DEM	GeoTIFF (Raster)	10m – 30m	Derive elevation, slope, aspect; supports hazard interpretation and accessibility analysis
Population Exposure	Population counts, households, population density	Department of Census & Statistics (www.statistics.gov.lk)	CSV / Vector (Tabular)	GN Division (preferred)	Measures number of people and households exposed to hazards
Population Vulnerability	Elderly (65+), children (0–14), disabled persons, poverty / low-income indicators	Department of Census & Statistics (www.statistics.gov.lk) / Ministry of Health (www.health.gov.lk)	CSV / Vector (Tabular)	GN Division (preferred)	Calculates vulnerability index to prioritize high-risk communities
Hospital Accessibility	Hospital locations, bed capacity, ICU beds, emergency services	Ministry of Health (www.health.gov.lk)	CSV / GeoJSON (Points)	Point level (facility based)	Evaluate medical response capability and accessibility to affected areas
Evacuation Shelters	Shelter locations, maximum capacity, current occupancy (if available), facilities (water, toilets, power, etc.)	Disaster Management Centre (DMC) (www.dmc.gov.lk) / District Disaster Management Units (DDMCUs)	CSV / GeoJSON (Points)	Point level (facility based)	Identify suitable evacuation locations and available shelter capacity
Road Network & Accessibility	Road network, road classification, bridges, road condition / accessibility	Road Development Authority (RDA) (www.rda.gov.lk) / Survey Department / OpenStreetMap	Shapefile / GeoJSON (Line)	Road-segment level	Analyze accessibility for evacuation routes and emergency response
Administrative Boundaries	District, DS Division, GN Division boundaries	Survey Department of Sri Lanka (www.survey.gov.lk) / Department of Census & Statistics	Shapefile / GeoJSON (Polygon)	Administrative level	Standardizes all datasets to the same spatial reference for integration
Historical Flood & Landslide Events	Historical events (date, location, hazard type, deaths, injuries, displaced people, damages)	Disaster Management Centre (DMC) (www.dmc.gov.lk) / NBRO / EM-DAT (www.emdat.be)	CSV / GeoJSON (Polygon/Point)	District / GN Division / Event based	Validate decision support system using past disaster events and impacts
Land Use / Land Cover (Optional)	Land use / land cover classification (residential, agriculture, forest, water, industrial, etc.)	Survey Department of Sri Lanka / ESA WorldCover / Copernicus (https://worldcover2021.esa.int)	Raster / Vector	10m – 30m	Estimate exposure of different land uses and support impact assessment

Emergency Response and Resource Coordination

Factor	Dataset Needed	Where to Request in Sri Lanka	Format	Resolution	Purpose
Ambulance Fleet Counts & Station, Hospital Locations	Ambulance fleet counts, station and hospital locations	Ministry of Health (health.gov.lk); Suwaseriya Foundation (1990 Ambulance Service); District hospital directories via Ministry of Health	Vector / tabular (point)	Facility-level	Emergency response capacity mapping and response-time estimation
Shelter / Safety Center Locations & Capacity	Registered shelter locations and capacity figures	Disaster Management Centre (dmc.gov.lk); District/Divisional Secretariat records	Vector / tabular (point, capacity attribute)	Facility-level	Evacuation planning and shelter allocation optimization
District / Divisional Administrative Boundaries	Administrative boundary layers	Survey Department (survey.gov.lk); Department of Census & Statistics (statistics.gov.lk)	Vector (polygon)	District / DS division level	Spatial aggregation unit for combining all component datasets and reporting
Relief Stock / NFI Availability Patterns	Relief stock and non-food item (NFI) inventory records	Disaster Management Centre (dmc.gov.lk); National Disaster Relief Services Centre (NDRSC)	Tabular / time-series	Warehouse / district level	Supply chain optimization and relief stock forecasting
Full Road Graph	Complete road network graph	Survey Department (survey.gov.lk); OpenStreetMap / HOT Export (export.hotosm.org)	Vector (line, network graph)	Road-segment level	Routing algorithm input (VRP/ILP) and travel-time modeling
Road / Bridge Damage Counts by District	Damage counts for roads and bridges	Road Development Authority – RDA (rda.gov.lk); Disaster Management Centre situation reports	Tabular / vector	District level	Identifies blocked or damaged routes affecting logistics planning
Flood-Blocked Road Segments (Derived)	Flood-affected road segments	Derived internally: Road graph overlaid with flood extent maps (Irrigation Department; Disaster Management Centre)	Vector (line, derived)	Road-segment level	Dynamic routing avoidance during active flood events
Road Classification (Optional)	Road class/hierarchy attributes	Survey Department (survey.gov.lk); Road Development Authority (rda.gov.lk)	Vector attribute (line)	Road-segment level	Prioritizes primary/secondary roads for routing weight
District / GN-Division Population & Age Structure (Derived)	Population counts and age structure	Department of Census & Statistics (statistics.gov.lk)	Tabular	Grama Niladhari division level	Vulnerability-weighted resource allocation (elderly/children prioritization)
Gridded Population	Gridded population estimates	WorldPop (worldpop.org); Department of Census & Statistics	Raster	100m grid	Fine-grained population distribution for demand estimation
Event-Level Disaster Records (Fatalities, Displaced, Duration)	Historical disaster event records	Disaster Management Centre – DesInventar Sri Lanka (desinventar.net, country: LKA)	Tabular	Event-level	Historical severity benchmarking and model training data
Named-Event Situation / PDNA Reports	Situation reports and Post-Disaster Needs Assessment (PDNA) documents	Disaster Management Centre situation reports; Ministry of Disaster Management / UNDP PDNA reports	PDF / document	Event-level	Qualitative validation and cross-reference for damage assessment